Exercise 1: SQL Fundamentals.

employee_db

ID	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1) SELECT Statement

Write a SQL query to retrieve all columns from the employees table:

```
SELECT *
FROM employees;
```

2) SELECT DISTINCT Statement

Write a SQL query to find all the unique departments in the employees table.

```
SELECT DISTINCT department
FROM employees;
```

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3) ORDER BY Statement

Write a SQL query to retrieve all employees first and last names, ordered by salary in descending order.

```
SELECT first_name;
       last_name,
       salary
FROM employees
ORDER BY salary DESC;
```

First-name	Last-name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith.	48000

4) LIMIT Statement

Write a SQL to retrieve the top 5 highest-paid employees

```
SELECT*
FROM employees
ORDERBY salary DESC
LIMIT 5 ;
```

	First_name	Last_name	Salary
1)	Emily	Davis	62000
2)	Mike	Johnson	60000
3)	Robert	Wilson	59000
4)	John	Doe	55000
5)	Sarah	Brown	53000

5) WHERE Statement

Write a SQL query to find employees who work in the IT department.

```
SELECT department, first_name, last_name.
FROM employees.
WHERE department='IT';
```

First_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6) AND Statement

Write a SQL query to find employees who work in the finance department AND have a salary greater than 58000.

```
SELECT *
FROM employees
WHERE department = 'Finance'
AND salary > 58000;
```

ID	first_name	last_name	department	salary	hire_date	city
3	Mike	Johnson	Finance	60000	2017-09-30	Los-Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston

7) OR Statement

Write a SQL query to find employees who work in the HR department OR the Marketing department

```
SELECT employee_id, first_name, last_name, department
FROM employees
WHERE department='HR' OR department = 'Marketing';
```

ID	First_name	Last_name	Department
2	Jane	Smith	HR
8	Jessica	Moore	HR
5	David	White	Marketing
9	Daniel	Clark	Marketing

8) NOT Statement

Write a SQL query to find employee who donot work in the IT department.

```
SELECT employee_ID,
       first_name,
       last_name,
       department
FROM employees
WHERE NOT department = 'IT';
```

ID	first_name	last_name	Department
2	Jane	Smith	HR
3	Mike	Johnsone	Finance
5	David	White	Marketing
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing

9) IN Statement

Write a SQL query to find employees who are in the HR, IT and Finance departments.

```
SELECT employee_ID,
       first_name,
       last_name,
       department
FROM employees
WHERE department IN ('HR','IT','Finance');
```

10) Combining Conditions

Write a SQL query to find employees who are in the IT department, have salary greater than 50000 and are located in New York

```
SELECT *
FROM employees
WHERE department = 'IT'
  AND salary > 50000
  AND city = 'New York';
```

---

11) Combining WHERE, AND and ORDER BY

Write a SQL query to retrieve the first and last name of employees who work in the Finance or Marketing department earn more than 52000, and order the results by salary in descending order.

```
SELECT first_name,
       last_name
FROM employees
WHERE department = 'Finance' OR department = 'Marketing'
AND salary > 52000
ORDER BY salary DESC;
```

First_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

12) Combining SELECT DISTINCT, WHERE, and IN

Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR

```
SELECT DISTINCT city
 FROM employees
  WHERE department NOT IN ('IT','HR');
```

---

13) Combining WHERE, NOT, AND and ORDER BY

Write a SQL query to retrieve employees who are NOT in the Finance department, have a salary greater than 50000 and order the results by hire date in ascending.

```
SELECT *
FROM employees
WHERE NOT department ='Finance'
   AND salary >50000
ORDER BY hire_date ASC;
```

---

14) Combining WHERE, OR, IN and LIMIT

Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department

```
SELECT *
FROM employees
WHERE (City = 'Chicago' OR City = 'Los Angeles')
   AND department IN ('IT', 'Marketing')
LIMIT 3;
```

ID	Firstname	last-name	department	salary	hire-date	city
6	Emily	Davis	IT	62000	2015/02/14	Chicago
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago